Incompressible Stokes Flow Annulus Benchmark Using Underworld3

Abstract

We reproduce the annulus Stokes benchmark of Thieulot and Puckett [2018] (TP2018) using Underworld3 (UW3). The benchmark describes incompressible isoviscous Stokes flow in a cylindrical annulus with tangential velocity boundary conditions imposed on both the inner and outer boundaries, driven by radially directed gravity and analytically prescribed buoyancy forcing that generates convection cells parameterised by the harmonic number k. We investigate the accuracy and convergence behaviour of several mixed finite-element discretisations, including $P_1 \times P_0$, $P_1 \times P_1$, $P_2 \times P_1$, and $P_3 \times P_2$. Numerical solutions are evaluated against the analytical velocity and pressure fields using relative L_2-norm errors. The higher-order Taylor–Hood discretisations reproduce the expected convergence hierarchy for smooth Stokes solutions, with approximately third-order velocity and second-order pressure convergence for $P_2 \times P_1$, and fourth-order velocity and third-order pressure convergence for $P_3 \times P_2$. In contrast, lower-order discretisations exhibit reduced robustness, particularly in pressure recovery. These results demonstrate that UW3 accurately reproduces analytical Stokes solutions in curved geometries and provides a robust framework for future annulus geodynamics simulations.

1 Introduction

Accurate numerical solution of the incompressible Stokes equations is fundamental to modelling mantle convection, lithospheric deformation, subduction dynamics, and planetary evolution. Many geodynamic applications require simulations in curved geometries, including cylindrical annuli, spherical caps, and full spherical shells. These geometries introduce additional numerical challenges associated with curved boundaries, pressure nullspaces, coordinate-dependent forcing terms, and tangential boundary conditions.

Reproducing analytical solutions remains one of the most rigorous approaches for assessing the correctness of numerical implementations. Unlike code-to-code comparisons, analytical benchmarks permit direct quantification of discretisation errors and convergence rates. For geodynamic Stokes solvers, analytical benchmarks are particularly valuable because they simultaneously test velocity–pressure coupling, incompressibility preservation, boundary condition enforcement, and curved-domain representation [Kramer et al., 2021, Davies et al., 2022].

The annulus benchmark introduced by TP2018 provides an exact analytical solution for incompressible isoviscous Stokes flow in a cylindrical shell subjected to radial gravity. The benchmark was specifically designed for geodynamics applications and has been used in the verification of geodynamic codes such as ASPECT and ELEFANT. Unlike classical Cartesian manufactured solutions, the benchmark includes curved inner and outer boundaries, non-trivial pressure distributions along the boundaries, coordinate-dependent buoyancy forcing, tangential velocity boundary conditions, and controllable azimuthal complexity through the harmonic parameter k. These properties make the benchmark particularly suitable for assessing finite-element formulations intended for annulus and spherical-shell mantle convection problems.

UW3 is a next-generation geodynamics framework built around PETSc and symbolic problem specification in Python. Similar benchmark studies have recently been reproduced using other geodynamic codes [Gassmöller et al., 2024, Davies et al., 2022]. Here, we reproduce the TP2018 annulus benchmark to validate the curved-domain finite-element implementations in UW3.

In this study, we use the TP2018 annulus benchmark to evaluate the convergence and accuracy of several mixed finite-element discretisations implemented in UW3. Numerical results are compared against the analytical velocity and pressure fields using global L_2-norm errors and direct boundary pressure diagnostics. Particular attention is given to the convergence hierarchy of higher-order Taylor–Hood discretisations and the behaviour of lower-order and equal-order element pairs.

2 Governing Equations

We consider the steady incompressible Stokes equations:

$$-\nabla \cdot \boldsymbol{\sigma} = \rho \mathbf{g}, \qquad \nabla \cdot \mathbf{u} = 0,$$

where $\mathbf{u}$ is velocity, p is pressure, ρ is density, and $\mathbf{g}$ is gravitational acceleration. The Cauchy stress tensor is

$$\boldsymbol{\sigma} = -p\mathbf{I} + 2\eta \boldsymbol{\varepsilon}(\mathbf{u}),$$

with viscosity η and strain-rate tensor

$$\boldsymbol{\varepsilon}(\mathbf{u}) = \frac{1}{2}\left(\nabla \mathbf{u} + (\nabla \mathbf{u})^T\right).$$

The computational domain is a two-dimensional annulus bounded by inner radius $R_1 = 1$ and outer radius $R_2 = 2$. Gravity acts radially inward:

$$\mathbf{g} = -\mathbf{e}_r.$$

The benchmark assumes constant viscosity $\eta = 1$ and prescribes analytical velocity, pressure, and density fields satisfying the Stokes equations exactly.

3 Analytical Benchmark Solution

Following Thieulot and Puckett [2018], the analytical velocity field in polar coordinates is defined as

$$v_\theta(r,\theta) = f(r)\cos(k\theta), \qquad v_r(r,\theta) = kg(r)\sin(k\theta),$$

where

$$f(r) = Ar + \frac{B}{r}, \qquad g(r) = \frac{A}{2}r + \frac{B}{r}\ln r + \frac{C}{r}.$$

The constants A and B are chosen such that radial velocity vanishes at both annulus boundaries:

$$A = -C\frac{2(\ln R_1 - \ln R_2)}{R_2^2 \ln R_1 - R_1^2 \ln R_2}, \qquad B = -C\frac{R_2^2 - R_1^2}{R_2^2 \ln R_1 - R_1^2 \ln R_2}.$$

The pressure field is

$$p(r,\theta) = kh(r)\sin(k\theta) + \rho_0(R_2 - r),$$

where

$$h(r) = \frac{2g(r) - f(r)}{r}.$$

The corresponding density field is

$$\rho(r,\theta) = kM(r)\sin(k\theta) + \rho_0,$$

with

$$M(r) = g''(r) - \frac{g'(r)}{r} - \frac{g(r)}{r^2}(k^2-1) + \frac{f(r)}{r^2} + \frac{f'(r)}{r}.$$

The benchmark calculations use the standard parameter values

$$R_1 = 1, \qquad R_2 = 2, \qquad C = -1, \qquad \rho_0 = 0.$$

Increasing the harmonic parameter k increases azimuthal complexity and produces $2k$ convection cells in the annulus. This enables systematic evaluation of numerical performance across increasingly oscillatory flow structures.

4 Numerical Methodology

4.1 Underworld3 Formulation

The benchmark is implemented in Underworld3 using PETSc-based finite-element Stokes solvers. The governing equations are assembled symbolically using SymPy and solved through PETSc Krylov subspace methods.

A constant-pressure nullspace is supplied to PETSc to remove the pressure gauge ambiguity. Following solution, the domain-average numerical pressure is removed prior to error evaluation.

The annulus meshes are generated using unstructured triangular elements with progressively refined nominal cell sizes. The mesh geometry degree is set to linear for all reported calculations.

4.2 Finite-Element Discretisations

The following mixed finite-element pairs are evaluated:

Velocity space	Pressure space	Description
P_1	P_0	Linear velocity, discontinuous constant pressure
P_1	P_1	Equal-order continuous discretisation
P_2	P_1	Taylor–Hood discretisation
P_3	P_2	Higher-order Taylor–Hood discretisation

The quadrature degree is chosen according to the highest polynomial order present in the discretisation. For smooth incompressible Stokes solutions, stable Taylor–Hood-type discretisations are expected to exhibit velocity convergence of order $p+1$ and pressure convergence of order p for a $P_{p+1} \times P_p$ discretisation [Girault and Raviart, 1986, Boffi et al., 2013].

4.3 Error Metrics

Relative L_2-norm errors are computed as

$$\|e_q\|_{L_2,\mathrm{rel}} = \left(\frac{\int_\Omega |q_h - q|^2 \, d\Omega}{\int_\Omega |q|^2 \, d\Omega} \right)^{1/2},$$

where q_h is the numerical solution and q is the analytical solution. Convergence rates are computed from successive mesh refinements.

5 Results

5.1 Analytical Benchmark Fields

Figure 1 illustrates the analytical density, velocity, and pressure fields for several harmonic values k. Increasing k produces increasingly complex azimuthal structure while preserving the exact incompressible solution. The benchmark simultaneously tests curved boundary representation, velocity–pressure coupling, radial gravity forcing, tangential slip-compatible velocity fields, and pressure recovery on curved boundaries.

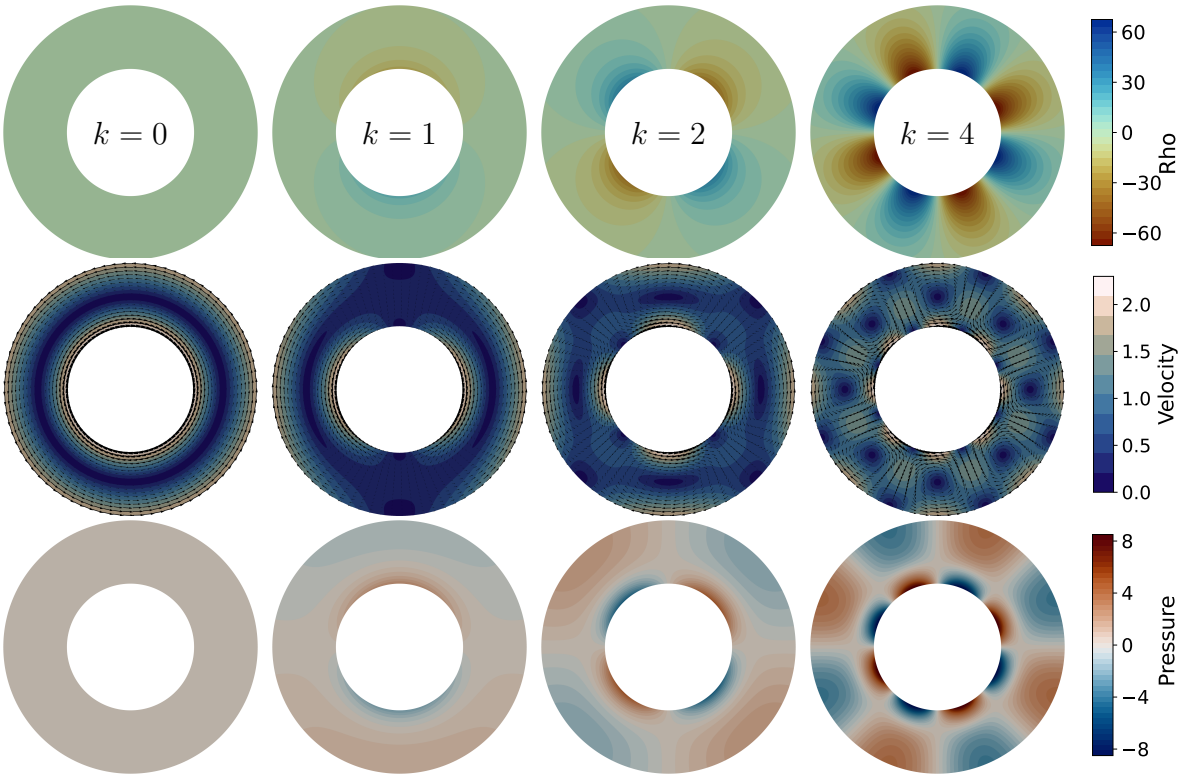

Figure 1: Analytical density, velocity, and pressure fields for the Thieulot annulus benchmark. Columns correspond to increasing k, which increases the number of convection cells in the annulus.

5.2 Convergence Behaviour

The convergence behaviour for all tested discretisations is shown in Fig. 2. The $P_2 \times P_1$ Taylor–Hood discretisation exhibits approximately third-order velocity convergence and second-order pressure convergence, consistent with theoretical expectations for smooth Stokes solutions.

The higher-order $P_3 \times P_2$ discretisation achieves approximately fourth-order velocity convergence and third-order pressure convergence on coarse-to-intermediate mesh resolutions. At the finest resolutions, convergence rates begin to plateau, likely reflecting accumulated effects from geometry approximation, quadrature errors, and solver tolerances.

The lower-order discretisations show reduced robustness. The $P_1 \times P_0$ pair fails to produce reliable convergence behaviour for this benchmark, while the equal-order $P_1 \times P_1$ discretisation produces acceptable velocity convergence but weaker and less consistent pressure convergence. These observations are consistent with established finite-element theory for incompressible Stokes flow [Boffi et al., 2013, Thieulot and Bangerth, 2022].

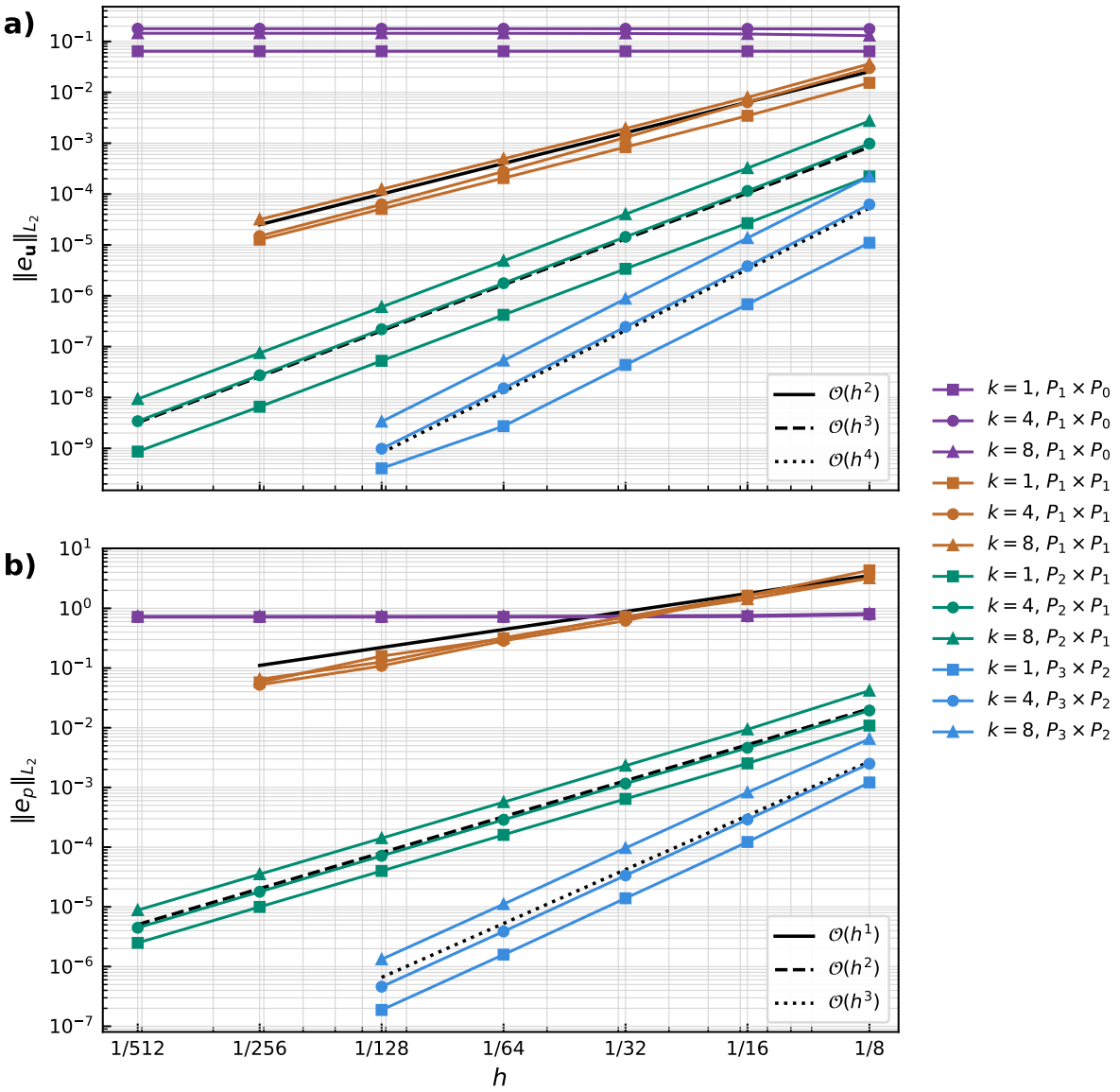

Figure 2: Convergence of relative volume L_2-norm velocity and pressure errors for the annulus Thieulot benchmark. Reference lines indicate the expected smooth-solution rates for the higher-order element pairs.

Table 1: Annulus Thieulot convergence for $P_1 \times P_0$.

	$k = 1$				$k = 4$				$k = 8$			
h	$\lvert e_v \rvert_2$	rate	$\lvert e_p \rvert_2$	rate	$\lvert e_v \rvert_2$	rate	$\lvert e_p \rvert_2$	rate	$\lvert e_v \rvert_2$	rate	$\lvert e_p \rvert_2$	rate
1/8	6.398e-02	–	8.040e-01	–	1.763e-01	–	7.800e-01	–	1.298e-01	–	8.163e-01	–
1/16	6.427e-02	-0.01	7.384e-01	0.12	1.775e-01	-0.01	7.295e-01	0.10	1.401e-01	-0.11	7.589e-01	0.11
1/32	6.426e-02	0.00	7.220e-01	0.03	1.780e-01	-0.00	7.161e-01	0.03	1.435e-01	-0.04	7.403e-01	0.04
1/64	6.426e-02	0.00	7.177e-01	0.01	1.781e-01	-0.00	7.124e-01	0.01	1.445e-01	-0.01	7.349e-01	0.01
1/128	6.425e-02	0.00	7.167e-01	0.00	1.782e-01	-0.00	7.114e-01	0.00	1.447e-01	-0.00	7.334e-01	0.00
1/256	6.425e-02	0.00	7.164e-01	0.00	1.782e-01	-0.00	7.112e-01	0.00	1.448e-01	-0.00	7.330e-01	0.00
1/512	6.425e-02	0.00	7.163e-01	0.00	1.782e-01	-0.00	7.111e-01	0.00	1.448e-01	-0.00	7.329e-01	0.00

Table 2: Annulus Thieulot convergence for $P_1 \times P_1$.

	$k = 1$				$k = 4$				$k = 8$			
h	$\lvert e_v \rvert_2$	rate	$\lvert e_p \rvert_2$	rate	$\lvert e_v \rvert_2$	rate	$\lvert e_p \rvert_2$	rate	$\lvert e_v \rvert_2$	rate	$\lvert e_p \rvert_2$	rate
1/8	1.537e-02	–	4.325e+00	–	2.994e-02	–	3.202e+00	–	3.629e-02	–	3.184e+00	–
1/16	3.459e-03	2.15	1.628e+00	1.41	6.385e-03	2.23	1.625e+00	0.98	7.925e-03	2.19	1.417e+00	1.17
1/32	8.358e-04	2.05	7.174e-01	1.18	1.293e-03	2.30	6.195e-01	1.39	1.937e-03	2.03	7.144e-01	0.99
1/64	2.056e-04	2.02	3.115e-01	1.20	2.776e-04	2.22	2.855e-01	1.12	4.931e-04	1.97	3.210e-01	1.15
1/128	5.095e-05	2.01	1.582e-01	0.98	6.252e-05	2.15	1.083e-01	1.40	1.243e-04	1.99	1.260e-01	1.35
1/256	1.266e-05	2.01	5.740e-02	1.46	1.487e-05	2.07	5.231e-02	1.05	3.158e-05	1.98	6.476e-02	0.96

Table 3: Annulus Thieulot convergence for $P_2 \times P_1$.

	$k = 1$				$k = 4$				$k = 8$			
h	$\lvert e_v \rvert_2$	rate	$\lvert e_p \rvert_2$	rate	$\lvert e_v \rvert_2$	rate	$\lvert e_p \rvert_2$	rate	$\lvert e_v \rvert_2$	rate	$\lvert e_p \rvert_2$	rate
1/8	2.218e-04	–	1.087e-02	–	9.781e-04	–	1.944e-02	–	2.731e-03	–	4.174e-02	–
1/16	2.675e-05	3.05	2.530e-03	2.10	1.151e-04	3.09	4.611e-03	2.08	3.225e-04	3.08	9.354e-03	2.16
1/32	3.388e-06	2.98	6.427e-04	1.98	1.437e-05	3.00	1.159e-03	1.99	4.016e-05	3.01	2.310e-03	2.02
1/64	4.205e-07	3.01	1.601e-04	2.01	1.763e-06	3.03	2.869e-04	2.01	4.850e-06	3.05	5.666e-04	2.03
1/128	5.254e-08	3.00	3.992e-05	2.00	2.190e-07	3.01	7.152e-05	2.00	6.001e-07	3.01	1.412e-04	2.00
1/256	6.575e-09	3.00	9.979e-06	2.00	2.730e-08	3.00	1.787e-05	2.00	7.449e-08	3.01	3.524e-05	2.00
1/512	8.632e-10	2.93	2.494e-06	2.00	3.412e-09	3.00	4.466e-06	2.00	9.287e-09	3.00	8.802e-06	2.00

Table 4: Annulus Thieulot convergence for $P_3 \times P_2$.

	$k = 1$				$k = 4$				$k = 8$			
h	$\lvert e_v \rvert_2$	rate	$\lvert e_p \rvert_2$	rate	$\lvert e_v \rvert_2$	rate	$\lvert e_p \rvert_2$	rate	$\lvert e_v \rvert_2$	rate	$\lvert e_p \rvert_2$	rate
1/8	1.102e-05	–	1.210e-03	–	6.233e-05	–	2.498e-03	–	2.198e-04	–	6.495e-03	–
1/16	6.792e-07	4.02	1.220e-04	3.31	3.811e-06	4.03	2.905e-04	3.10	1.363e-05	4.01	8.249e-04	2.98
1/32	4.360e-08	3.96	1.392e-05	3.13	2.441e-07	3.96	3.358e-05	3.11	8.723e-07	3.97	9.664e-05	3.09
1/64	2.729e-09	4.00	1.585e-06	3.13	1.510e-08	4.01	3.870e-06	3.12	5.323e-08	4.03	1.110e-05	3.12
1/128	4.043e-10	2.75	1.877e-07	3.08	9.826e-10	3.94	4.601e-07	3.07	3.344e-09	3.99	1.322e-06	3.07

5.3 Boundary Pressure Recovery

Pressure profiles extracted along the inner and outer annulus boundaries are compared against analytical solutions in Fig. 3. The higher-order Taylor–Hood discretisations reproduce the analytical pressure distributions accurately across the annulus boundaries, including extrema near regions of high azimuthal variation. Pressure discrepancies are most pronounced for lower-order discretisations and become increasingly visible for larger k, where sharper angular pressure gradients must be resolved.

Because the benchmark includes non-trivial pressure variation directly on curved boundaries, these comparisons provide a stringent test of both pressure interpolation quality and curved-domain numerical accuracy.

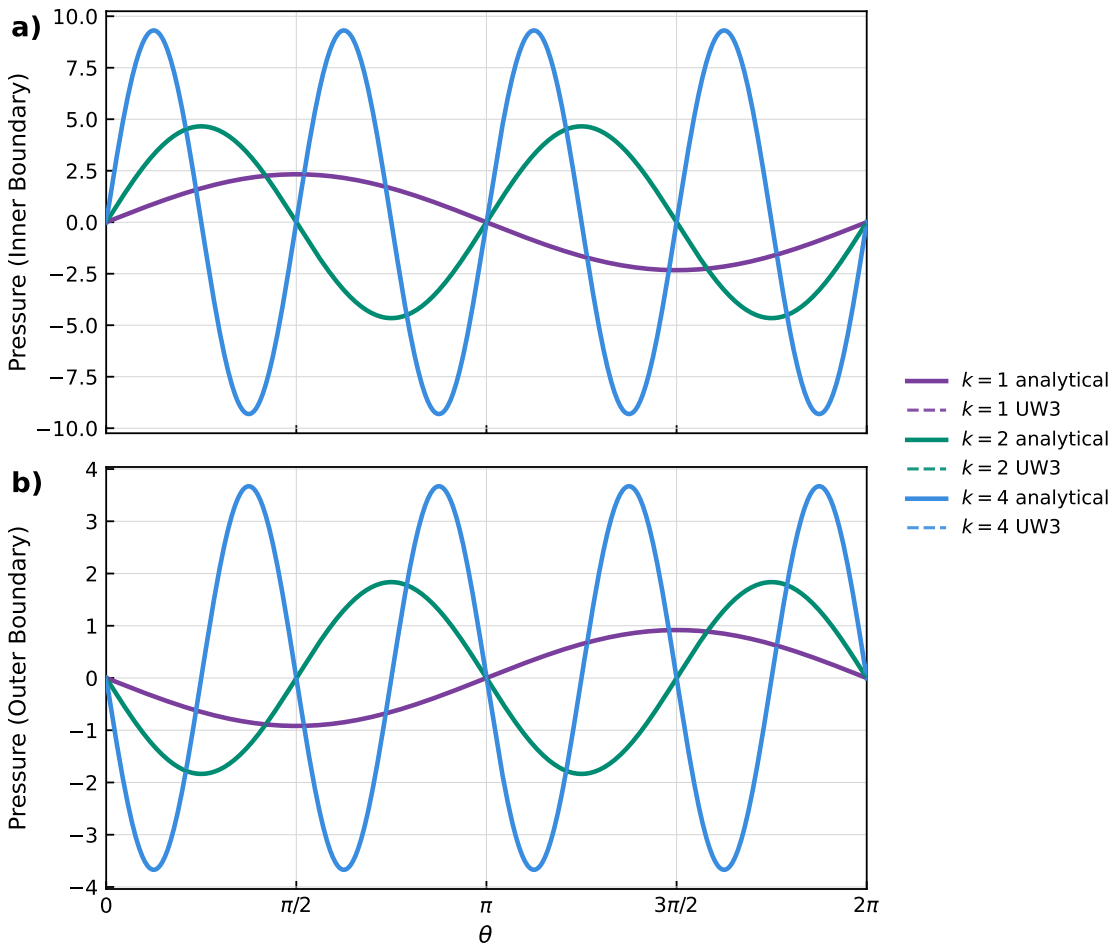

Figure 3: Pressure along the inner and outer annulus boundaries for $k = 1, 2, 4$, using the $P_2 \times P_1$, $h = 1/128$ calculations. Solid curves denote the analytical solution and dashed curves denote UW3 values sampled from the numerical pressure field.

5.4 Implications for Spherical Geodynamics

Although the benchmark is formulated in a two-dimensional annulus, its numerical characteristics are directly relevant to spherical-shell mantle convection calculations. In particular, the benchmark tests curved-boundary finite-element behaviour, radial body-force implementation, pressure nullspace treatment, incompressibility preservation, and higher-order mixed discretisations. These are all essential ingredients in regional and global spherical geodynamics models.

The strong convergence behaviour observed for higher-order Taylor–Hood discretisations therefore provides confidence in Underworld3 implementations intended for annulus and spherical-shell mantle convection applications.

6 Discussion

The present results demonstrate that Underworld3 reproduces the analytical annulus benchmark solution with convergence behaviour consistent with finite-element theory. The benchmark additionally highlights several important numerical considerations.

First, higher-order mixed discretisations provide substantially improved pressure accuracy relative to lower-order formulations. This is particularly important in geodynamics applications where pressure gradients strongly influence buoyancy-driven flow and dynamic topography.

Second, curved geometries remain sensitive to geometric approximation order. Although linear geometry representation was sufficient to recover theoretical convergence trends for the present benchmark, higher-order curved geometry representations may become increasingly important for extremely fine meshes or strongly curved spherical-shell problems.

Third, equal-order discretisations continue to exhibit weaker pressure robustness relative to stable Taylor–Hood formulations. This behaviour aligns with expectations from the Ladyzhenskaya–Babuska–Brezzi stability condition.

The benchmark therefore provides a valuable verification framework not only for Underworld3, but also for future development of curved-element geometries, advanced boundary-condition implementations, and spherical-shell finite-element formulations.

7 Conclusions

We have presented a verification study of the Thieulot–Puckett annulus Stokes benchmark using Underworld3. The benchmark provides a stringent test of incompressible Stokes flow in curved geometries through the combination of radial gravity, coordinate-dependent forcing, tangential boundary velocity conditions, and non-trivial pressure variation on curved annulus boundaries.

The principal findings are:

1. Higher-order Taylor–Hood discretisations reproduce the expected smooth-solution convergence hierarchy.

2. The $P_2 \times P_1$ discretisation achieves approximately third-order velocity and second-order pressure convergence.

3. The $P_3 \times P_2$ discretisation achieves approximately fourth-order velocity and third-order pressure convergence before fine-resolution error saturation.

4. Lower-order discretisations exhibit weaker pressure robustness and reduced convergence quality.

5. Underworld3 accurately reproduces analytical Stokes solutions in curved annular geometries.

These results establish the Thieulot–Puckett annulus benchmark as an effective verification framework for Underworld3 annulus and spherical-shell geodynamics applications.

References

Daniele Boffi, Franco Brezzi, and Michel Fortin. *Mixed Finite Element Methods and Applications*. Springer Series in Computational Mathematics. Springer Berlin Heidelberg, 2013. ISBN 978-3-642-36519-5. doi: 10.1007/978-3-642-36519-5.

D. Rhodri Davies, Stephan C. Kramer, Sia Ghelichkhan, and Angus Gibson. Towards automatic finite-element methods for geodynamics via firedrake. *Geoscientific Model Development*, 15 (13):5127–5166, 2022. doi: 10.5194/gmd-15-5127-2022.

Rene Gassmöller, Juliane Dannberg, Wolfgang Bangerth, Elbridge Gerry Puckett, and Cedric Thieulot. Benchmarking the accuracy of higher-order particle methods in geodynamic models of transient flow. *Geoscientific Model Development*, 17(10):4115–4134, 2024. doi: 10.5194/gmd-17-4115-2024.

Vivette Girault and Pierre-Arnaud Raviart. *Finite Element Methods for Navier–Stokes Equations*. Springer Series in Computational Mathematics. Springer Berlin Heidelberg, 1986. ISBN 978-3-642-61623-5. doi: 10.1007/978-3-642-61623-5.

Stephan C. Kramer, D. Rhodri Davies, and Cian R. Wilson. Analytical solutions for mantle flow in cylindrical and spherical shells. *Geoscientific Model Development*, 14(4):1899–1919, 2021. doi: 10.5194/gmd-14-1899-2021.

C. Thieulot and E. G. Puckett. Incompressible stokes flow in an annulus: An analytical solution and numerical benchmark. Preprint submitted to Computers & Geosciences, 2018. URL `https://www.math.ucdavis.edu/~egp/PUBLICATIONS/JOURNAL_ARTICLES/SUBMITTED/CAPT-EGP-2018.pdf`.

Cedric Thieulot and Wolfgang Bangerth. On the choice of finite element for applications in geodynamics. *Solid Earth*, 13(1):229–249, 2022. doi: 10.5194/se-13-229-2022.